Lab 3 Report Microprocessor

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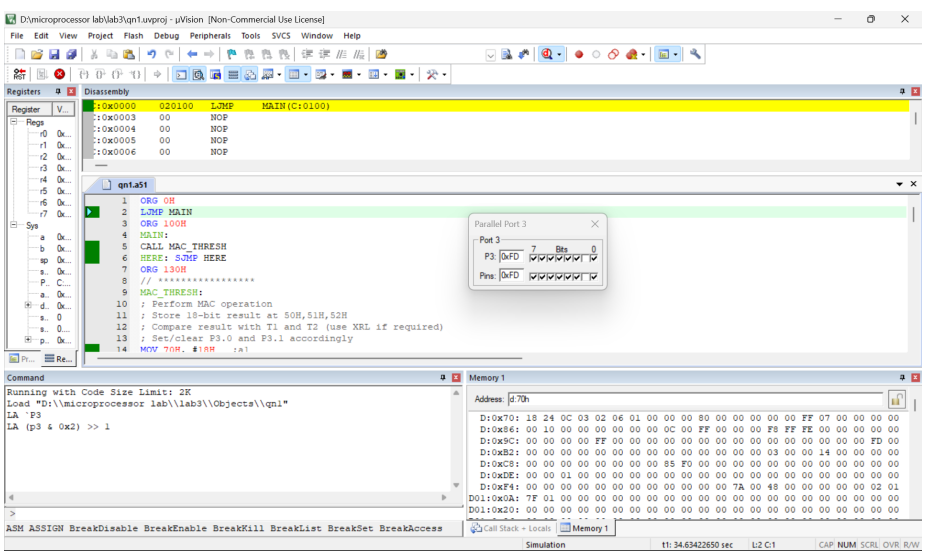
1 Problem 1

1.1 Objective

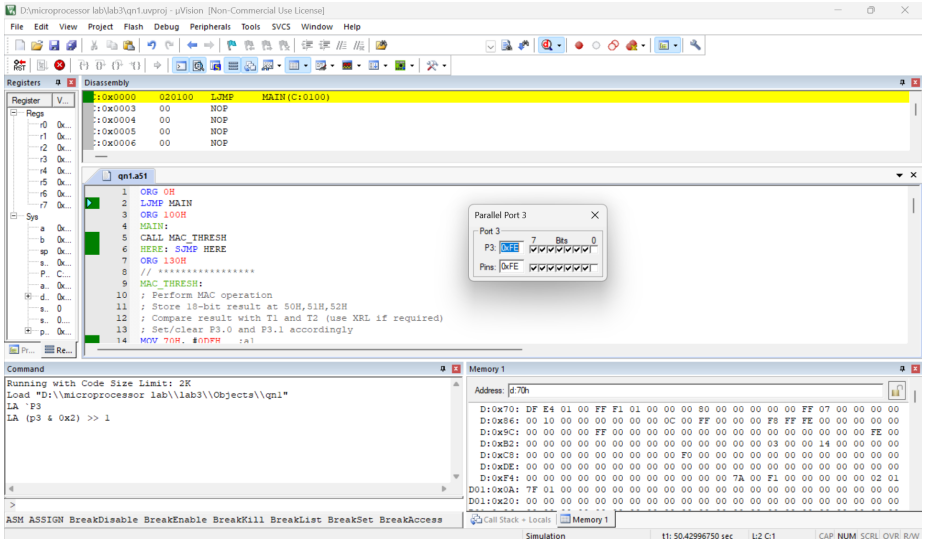
The primary goal of this experiment is to implement a Multiply and Accumulate (MAC) engine using 8051 Assembly language. This involves performing multi-byte arithmetic to handle products that exceed 8 bits and implementing a dual-threshold decision logic. And then giving the desired output by comparing with the provided number.

1.2 Simulation

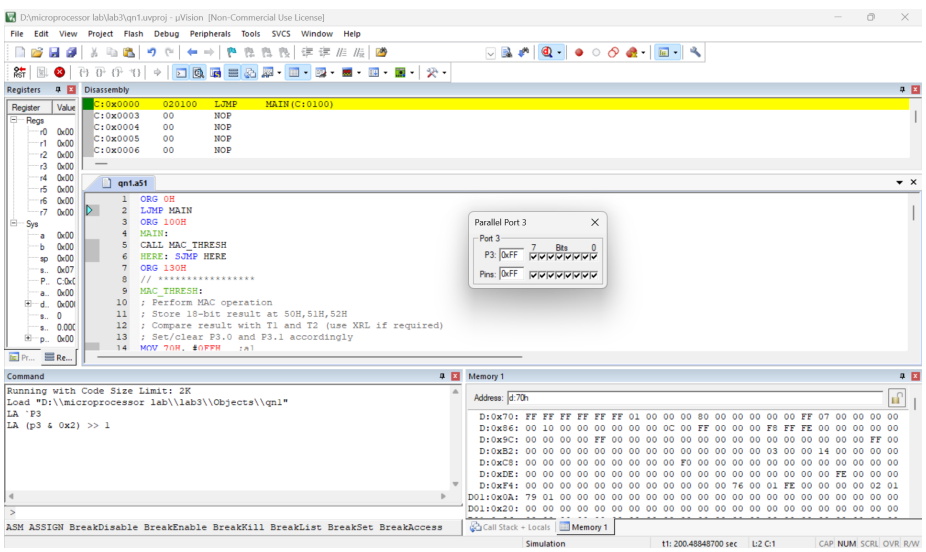
1.2.1 Test case 1 for a1,a2,a3 = 18H,24H,0CH & b1,b2,b3 = 03H,02H,06H



1.2.2 Test case 2 for a1,a2,a3 = DFH,E4H,01H & b1,b2,b3 = 00H,FFH,F1H



1.2.3 Test case 2 for a1,a2,a3 = FFH,FFH,FFH & b1,b2,b3 = FFH,FFH,FFH



1.3 Observation

- Precision Management: Since the product of two 8-bit numbers results in a 16-bit value, and the sum of three such products can exceed 16 bits (up to 18 bits), the use of a 3-byte accumulator (locations 50H–52H) is necessary to prevent overflow
- Subtraction-based Comparison: The CJNE (Compare and Jump if Not Equal) instruction, combined with the Carry flag (CY), effectively acts as a "less than or equal to" evaluator across multiple bytes
- Logic Verification:For small inputs (Case 1), x remains below $T2$, resulting in P3.0=1, P3.1=0.For maximum inputs (Case 3: all FFH), the MAC result greatly exceeds $T1$, triggering P3.0=1, P3.1=1

1.4 Conclusion

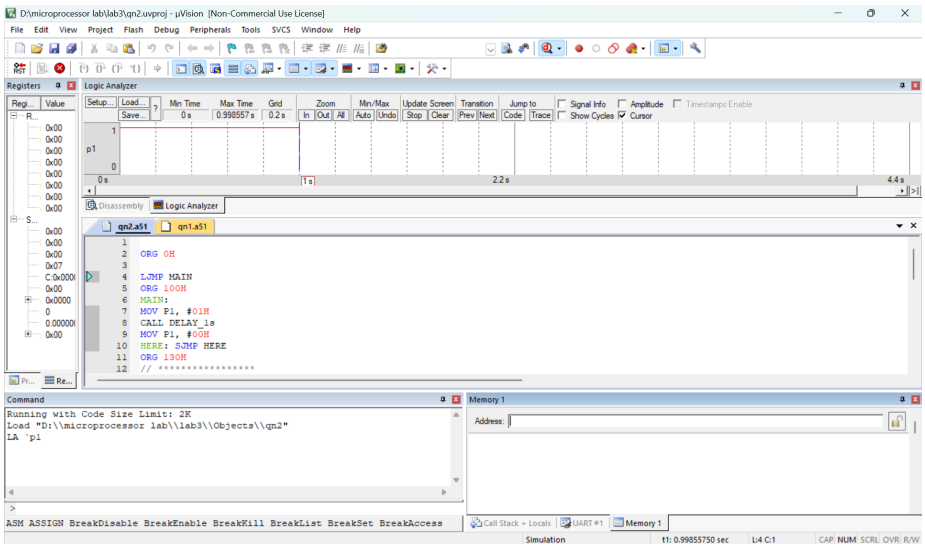
The experiment successfully demonstrates that a standard 8-bit microcontroller can perform complex DSP tasks like MAC and thresholding through software implementation of multi-precision arithmetic. By cascading the ADDC instruction, we successfully computed an 18-bit result. The conditional jumps based on the Carry flag allowed for precise decision-making across three distinct ranges defined by $T1$ and $T2$.

2 Problem 2

2.1 Objective

The objective of this experiment is to create a precise time delay of 1 second using nested subroutine calls in 8051 Assembly. It focuses on understanding the relationship between the machine cycle frequency (24MHz), instruction execution time, and loop iterations to achieve controlled timing in embedded systems.

2.2 Simulations



2.3 Observations

- Logic Analyzer Output: When monitoring Port 1.0, the signal stays "High" (1) for exactly 1.0 seconds before transitioning to "Low" (0).
- Clock Frequency Impact: At 24MHz, the machine cycle frequency is $24MHz/12 = 2MHz$. This means one machine cycle takes $0.5\mu s$.
- Subroutine Granularity: * The delay_250us uses a DJNZ loop (2 cycles $\times 244 iterations = 488$ cycles). Including PUSH, MOV, and POP, it totals roughly 500 cycles. $500 \times 0.5\mu s = 250\mu s$.The delay_10ms calls this 40 times ($40 \times 250\mu s = 10ms$).The DELAY_1s calls the 10ms block 100 times, resulting in a 1s duration.

2.4 Conclusion

The experiment demonstrates that high-level delays (seconds) can be accurately constructed by nesting lower-level delays (microseconds and milliseconds). By using the DJNZ instruction and carefully calculated loop counters, we can compensate for the high execution speed of the 8051. The accuracy of the delay is directly dependent on the crystal frequency; any change in the 24MHz clock would scale the delay proportionally.